

A Level-Set Framework for Multilayer Aircraft Ice Accretion

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Abstract

This work presents a continuous level-set methodology for the prediction of multilayer aircraft ice accretion on unstructured meshes. Conventional multilayer icing simulations commonly update the ice geometry through explicit displacement of surface nodes, which can become sensitive to mesh quality as strongly curved and interacting ice features develop. The proposed methodology instead represents the evolving ice–air interface implicitly as the zero contour of a signed-distance level-set function. To the authors’ knowledge, this work presents the first application of a continuous level-set formulation to the prediction of three-dimensional multilayer ice-accretion geometries. Implemented within the CHAMPS icing framework, the methodology combines surface-quantity propagation, semi-implicit Rosenbrock–Wanner time integration, local interface reconstruction for ice-volume control, and conformal surface extraction using MMG. Preliminary results are presented for representative configurations from the AIAA Ice Prediction Workshops. For the two-dimensional IPW1 Case 241, the difference in maximum ice height relative to experiment decreases from 6.47% using one accretion layer to 0.70% using 20 layers. Application to the three-dimensional Case 362 produces ice shapes in good agreement with experimental measurements and within the IPW2 participant envelope while remaining robust throughout the multilayer process. The completed work will extend the assessment to additional IPW2 configurations and investigate the influence of more advanced surface-roughness and ice-density models on complex three-dimensional glaze-ice accretion.

Introduction

Aircraft ice accretion can significantly degrade aerodynamic performance by modifying the effective geometry of lifting surfaces [1]. Numerical icing simulations generally employ a quasi-steady multilayer approach in which aerodynamic, droplet-impingement, and thermodynamic solutions are successively recomputed as the ice geometry evolves [2].

Geometry evolution remains an important numerical challenge. Conventional methods typically displace surface nodes along prescribed growth directions [3]. Although computationally efficient, repeated displacement can lead to mesh distortion or entanglement as strongly curved or interacting ice features develop, particularly in three dimensions [4, 5].

Level-set methods represent evolving interfaces implicitly through a scalar field and provide a framework for tracking complex moving boundaries [6–8]. They have been applied to moving-interface problems and, more recently, to aircraft ice accretion [9–12].

The present work develops a continuous level-set framework for two- and three-dimensional multilayer ice accretion within the CHAMPS solver [13]. The significant aspect of the proposed approach is the evolution of the ice interface independently of the connectivity and displacement of the aerodynamic surface mesh, followed by conformal extraction and remeshing between successive accretion layers. This formulation is intended to improve robustness for complex three-dimensional ice geometries and interacting ice fronts.

Methodology

The methodology is embedded within the quasi-steady icing framework shown in Fig. 1. For each layer, the aerodynamic flow, droplet impingement, and surface thermodynamics are computed sequentially. The resulting ice thickness drives the level-set evolution, after which the new surface is extracted and remeshed for the subsequent layer.

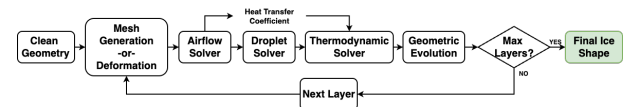


Figure 1: Quasi-steady multilayer ice-accretion framework implemented within CHAMPS.

Interface Representation and Advection

The ice–air interface is represented by a level-set function ϕ , with $\phi < 0$ in the ice, $\phi > 0$ in the fluid, and $\phi = 0$ at the interface, as illustrated in Fig. 2.

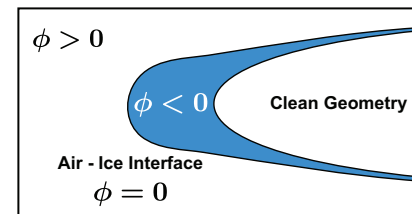


Figure 2: Level-set representation of the ice–air interface.

The initial signed-distance field, illustrated in Fig. 3(a), is obtained from the Eikonal equation [14],

$$|\nabla\phi| = 1, \quad \mathbf{n}_\phi = \frac{\nabla\phi}{|\nabla\phi|}, \quad (1)$$

where $\mathbf{n}_\phi$ denotes the unit normal vector to the level-set contours. The interface is advected using the level-set transport equation [7],

$$\frac{\partial\phi}{\partial t} + \mathbf{V}_{ice} \cdot \nabla\phi = 0, \quad (2)$$

where the ice-growth velocity is

$$\mathbf{V}_{ice} = \frac{h_{ice}}{\Delta t_{layer}} \mathbf{n}_\phi. \quad (3)$$

Here, h_{ice} is the local ice thickness and Δt_{layer} is the accretion-layer duration. As illustrated in Fig. 3(b), the surface ice thickness is propagated into the level-set domain along $\mathbf{n}_\phi$ prior to level-set advection.

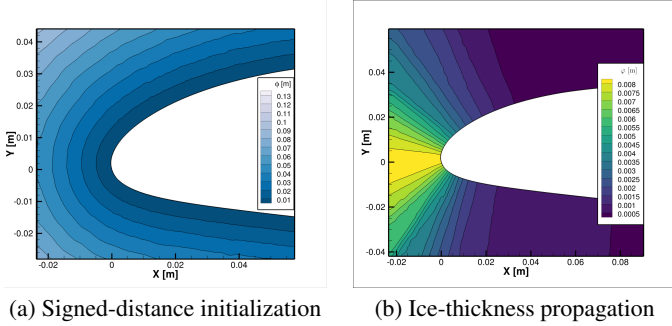


Figure 3: Level-set initialization and ice-thickness propagation prior to interface advection.

Time Integration and Surface Extraction

To alleviate restrictive time-step limitations associated with explicit level-set integration, the transport equation is advanced using a two-stage semi-implicit Rosenbrock–Wanner (ROW) scheme [15]. The time integration is written as

$$\phi^{n+1} = \phi^n + \sum_{j=1}^s m_j S_j, \quad (4)$$

where the stage increments S_i are obtained from

$$\left(\frac{\mathbf{I}}{\gamma \Delta t} + \mathbf{J} \right)^n S_i = -R \left(\phi^n + \sum_{j=1}^{i-1} a_{ij} S_j \right). \quad (5)$$

Here, $R(\phi)$ is the spatial residual, $\mathbf{J} = \partial R / \partial \phi$ its Jacobian, and S_i the stage increments. The resulting linear systems are solved using GMRES. The semi-implicit formulation improves stability and permits larger time steps on the locally refined auxiliary mesh used for level-set advection, illustrated in Fig. 4.

During advection, the ice volume is evaluated through local reconstruction of the zero level set within intersected elements. Advection is terminated when the geometric volume increment matches that predicted by the thermodynamic model. The $\phi = 0$ interface is then extracted using MMG [16] and remeshed for the subsequent accretion layer.

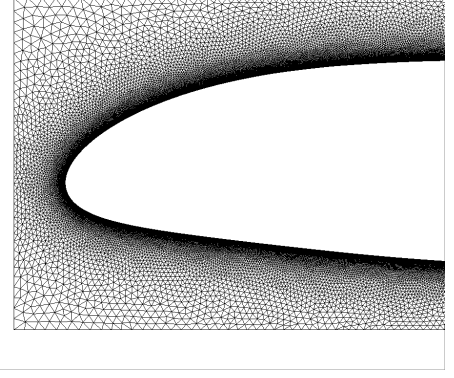


Figure 4: Adapted auxiliary mesh with local refinement in the ice-accretion region for level-set advection.

Preliminary Results

The methodology is first illustrated using a manufactured three-horn configuration designed to assess its ability to handle interacting fronts. As shown in Fig. 5, direct surface displacement can produce mesh entanglement as neighboring fronts interact, whereas the implicit level-set representation naturally accommodates their evolution and merging. The methodology is subsequently assessed on configurations from the AIAA Ice Prediction Workshops [17, 18].

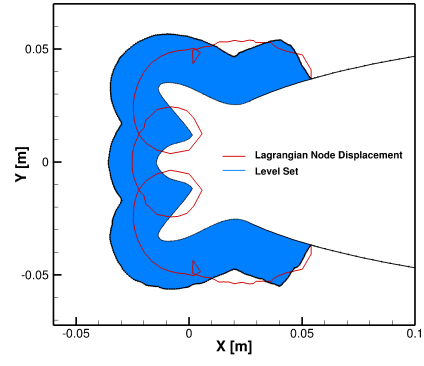


Figure 5: Evolution of interacting fronts using surface displacement and the level-set formulation.

Two-Dimensional Multilayer Convergence

IPW1 Case 241, corresponding to rime ice accretion on a NACA 23012 airfoil, is used to assess multilayer convergence [17]. Figure 6 compares predictions using 1, 10, and 20 layers with experiment.

The experimental maximum ice height is 10.87 mm. Predictions increase from 10.16 mm with one layer to 10.69 and 10.79 mm with 10 and 20 layers, reducing the relative difference from 6.47% to 1.68% and 0.70%, respectively. The limited change between 10 and 20 layers indicates that the ice geometry is approaching temporal convergence.

Three-Dimensional Multilayer Ice Accretion

The three-dimensional capability is assessed using Case 362, a NACA 0012 swept-wing configuration with a 36-in. chord and a 30° sweep angle, exhibiting spanwise-varying leading-edge ice characteristic of swept-wing icing configurations [19]. The level-set method remains robust throughout the multilayer process, successively ad-

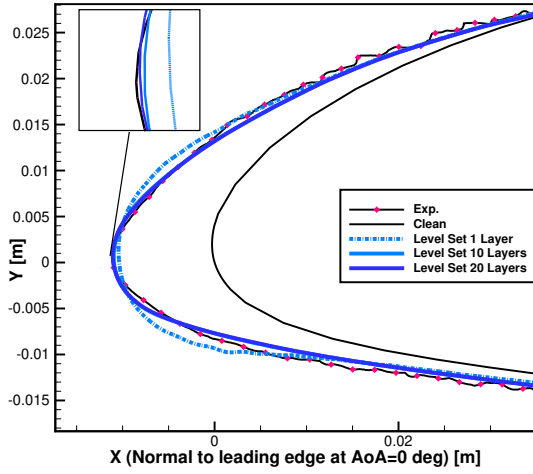


Figure 6: IPW1 Case 241 predictions using 1, 10, and 20 accretion layers.

vecting, extracting, and remeshing the three-dimensional interface. A three-dimensional view of the predicted ice geometry is shown in Fig. 7.

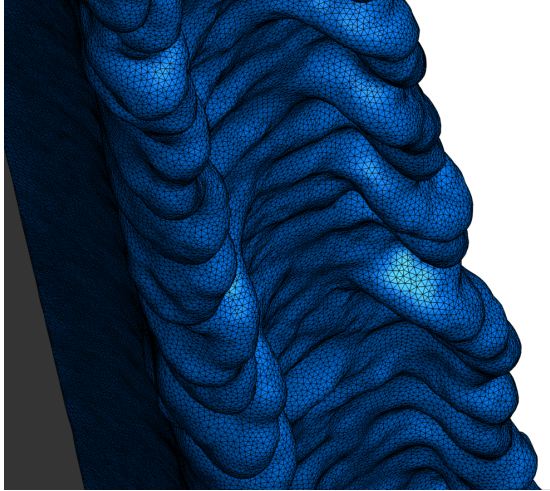


Figure 7: Three-dimensional view of the predicted ice geometry for Case 362.

The predicted ice geometry shows good agreement with the experimental ice shape and remains within the IPW2 participant envelope [18], as shown in Fig. 8.

Expected Results

The completed study will extend the assessment of the level-set methodology to additional configurations from IPW2, with emphasis on complex three-dimensional glaze-ice geometries. These cases will provide further evaluation of the robustness of the interface-propagation, extraction, and remeshing procedure over successive accretion layers. The influence of more advanced surface-roughness and ice-density models will also be investigated. The resulting predictions will be compared with available experimental measurements and workshop participant data to assess ice-shape agreement and multilayer sensitivity.

Conclusion

A continuous level-set methodology for two- and three-dimensional multilayer aircraft ice accretion has been developed within CHAMPS. Preliminary results demonstrate that the implicit interface representa-

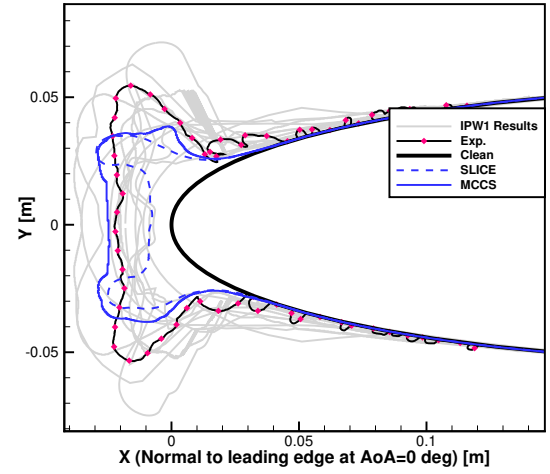


Figure 8: Comparison of the predicted Case 362 ice geometry with experimental and IPW2 participant data.

tion can evolve and remesh complex ice geometries without directly displacing the aerodynamic surface. For IPW1 Case 241, increasing the number of layers from 1 to 20 reduces the difference in maximum ice height relative to experiment from 6.47% to 0.70%. Application to the three-dimensional Case 362 produces ice geometries in good agreement with experiment and within the IPW2 participant envelope while remaining robust throughout the multilayer process. The planned extension to additional IPW2 configurations will further assess the methodology for complex three-dimensional glaze-ice accretion.

References

1. Y. Cao, W. Tan, and Z. Wu, "Aircraft icing: An ongoing threat to aviation safety," *Aerospace Science and Technology*, vol. 75, pp. 353–385, 2018.
2. R. Kind, M. Potapczuk, A. Feo, C. Golia, and A. Shah, "Experimental and computational simulation of in-flight icing phenomena," *Progress in Aerospace Sciences*, vol. 34, no. 5, pp. 257–345, 1998.
3. D. S. Thompson, X. Tong, Q. Arnoldus, E. Collins, E. A. Luke, and C. S. Bidwell, "Discrete surface evolution and mesh deformation for aircraft icing applications,"
4. J. Glimm, J. W. Grove, X. L. Li, and D. C. Tan, "Robust computational algorithms for dynamic interface tracking in three dimensions," *SIAM Journal on Scientific Computing*, vol. 21, no. 6, pp. 2240–2256, 2000.
5. I. Ozcer, D. Switchenko, G. S. Baruzzi, and J. Chen, "Multi-shot icing simulations with automatic re-meshing," in *SAE Technical Paper Series*, vol. 1 of *ICE*, SAE International, 2019.
6. S. Osher and J. A. Sethian, "Fronts propagating with curvature-dependent speed: Algorithms based on hamilton-jacobi formulations," *Journal of Computational Physics*, vol. 79, no. 1, pp. 12–49, 1988.
7. S. Osher and R. Fedkiw, *The Level Set Methods and Dynamic Implicit Surfaces*, vol. 57, pp. xiv+273. 05 2004.
8. F. Gibou, R. Fedkiw, and S. Osher, "A review of level-set methods and some recent applications," *Journal of Computational Physics*, vol. 353, pp. 82–109, 2018.
9. D. Pena, Y. Hoarau, and E. Laurendeau, "A single step ice accretion model using level-set method," *Journal of Fluids and Structures*, vol. 65, pp. 278–294, 2016.

10. S. Bourgault-Côté, J. Docampo-Sánchez, and E. Laurendeau, "Multilayer airfoil ice accretion simulations using a level-set method with b-spline representation," *AIAA Journal*, vol. 57, no. 8, pp. 3299–3308, 2019.
11. P. Lavoie, E. Radenac, G. Blanchard, E. Laurendeau, and P. Villedieu, "Immersed boundary methodology for multistep ice accretion using a level set," *Journal of Aircraft*, vol. 59, no. 4, pp. 912–926, 2022.
12. A. Donizetti, T. Bellosta, A. Rausa, B. Re, and A. Guardone, "Three-dimensional front-tracking technique for multistep simulations of aircraft icing," *Journal of Aircraft*, vol. 62, no. 2, pp. 328–340, 2025.
13. M. Parenteau, S. Bourgault-Cote, F. Plante, E. Kayraklioglu, and E. Laurendeau, "Development of parallel cfd applications with the chapel programming language,"
14. H. Xia and P. G. Tucker, "Finite volume distance field and its application to medial axis transforms," *International Journal for Numerical Methods in Engineering*, vol. 82, no. 1, pp. 114–134, 2010.
15. E. Hairer and G. Wanner, *Solving Ordinary Differential Equations II. Stiff and Differential-Algebraic Problems*, vol. 14 of *Springer Verlag Series in Comput. Math.* Springer Berlin, Heidelberg, 1996.
16. C. Dapogny, C. Dobrzynski, and P. Frey, "Three-dimensional adaptive domain remeshing, implicit domain meshing, and applications to free and moving boundary problems," *Journal of Computational Physics*, vol. 262, pp. 358–378, 2014.
17. E. Laurendeau, S. Bourgault-Cote, I. A. Ozcer, R. Hann, E. Radenac, and A. Pueyo, "Summary from the 1st aiaa ice prediction workshop," in *AIAA Aviation 2022 Forum*, p. 3398, American Institute of Aeronautics and Astronautics (AIAA), June 2022.
18. E. Laurendeau, M. Blanchet, M. K. Zayni, R. Hann, E. Radenac, I. Mussa, and A. Pueyo, "Summary from the 2nd aiaa ice prediction workshop," in *AIAA AVIATION FORUM AND ASCEND 2024*, American Institute of Aeronautics and Astronautics, July 2024.
19. J. M. Diebold, A. P. Broeren, and M. Bragg, "Aerodynamic classification of swept-wing ice accretion," in *5th AIAA Atmospheric and Space Environments Conference*, p. 2825, 2013.

h_{ice}	Local ice thickness	[m]
$\mathbf{n}_\phi$	Level-set interface normal	[−]
t	Time	[s]
Δt_{layer}	Accretion-layer duration	[s]
$\mathbf{V}_{ice}$	Ice-growth velocity	[m/s]

General symbols

Greek symbols

ϕ	Level-set signed-distance function	[m]
ρ_{ice}	Ice density	[kg/m ³]

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Definitions, Acronyms, Abbreviations

CFD	Computational Fluid Dynamics
IPW1	First AIAA Ice Prediction Workshop
IPW2	Second AIAA Ice Prediction Workshop
RANS	Reynolds-Averaged Navier–Stokes